SYLLABUS

Identification Theory

Students: Automation (AT) Bachelor
Year: Fourth
Semester: Seventh

Lecturer and Instructor:

Alexey Vedyakov	Associate Professor	vedyakov@itmo.ru
Alexey Ovcharov	Assistant Professor	a.ovcharov@itmo.ru

Course Description

The course gives basic knowledge parametric system identification, i.e. on estimating parameters of the mathematical models of systems. The course consists of two parts. The first part is devoted to offline identification, where the least square estimation method is considered in detail. The second part focuses on online identification, where standard gradient descent and novel dynamic regressor extension and mixing methods are explained. In the lab lessons, the students will work in MATLAB® and Simulink toolbox.

The course will run on September 28, 2026. The 40-hour course includes 16 hours of lectures and 24 hours of labs and consultations. The course ends with the final test.

Learning objectives

After the course, the students:

- know the basics of system identification, some offline and online identification methods,
- can estimate parameters of the linear regression models,
- can transform linear dynamic models to linear regression in order to estimate model parameters,
- can transform some nonlinear problems to linear regression,
- can compare quality of the identified models.

Materials

Main materials:

1. Lecture presentations
2. Lab Guides

Additional materials:

1. Petros Ioannou, Jing Sun. Robust Adaptive Control. Dover Publications. 2013.
2. Petros Ioannou, Bariş Fidan. Adaptive control tutorial. Society for Industrial and Applied Mathematics. 2006.
3. Shankar Sastry, Marc Bodson. Adaptive Control: Stability, Convergence, and Robustness. Prentice-Hall. 1989
4. Lennart Ljung. System Identification: Theory for the User. Pearson. 1998
5. A. Belov, S. Aranovskiy, R. Ortega, N. Barabanov and A. Bobtsov, "Enhanced Parameter Convergence for Linear Systems Identification: The DREM Approach," 2018 European Control Conference (ECC), 2018, pp. 2794-2799, doi: 10.23919/ECC.2018.8550338.

6. M. Korotina, S. Aranovskiy, R. Ushirobira and A. Vedyakov, "On parameter tuning and convergence properties of the DREM procedure," 2020 European Control Conference (ECC), 2020, pp. 53-58, doi: 10.23919/ECC51009.2020.9143808.
7. Aranovskiy, Stanislav, Alexey Bobtsov, Romeo Ortega, and Anton Pyrkin. "Parameters estimation via dynamic regressor extension and mixing." In 2016 American Control Conference (ACC), pp. 6971-6976. IEEE, 2016.
8. Ortega, Romeo, Stanislav Aranovskiy, Anton A. Pyrkin, Alessandro Astolfi, and Alexey A. Bobtsov. "New results on parameter estimation via dynamic regressor extension and mixing: Continuous and discrete-time cases." IEEE Transactions on Automatic Control 66, no. 5 (2020): 2265-2272.
9. Getting Started with Simulink https://www.mathworks.com/help/simulink/getting-started-with-simulink.html

Course policies

- The course may use various learning technologies to facilitate learning in traditional on-campus face-to-face education, such as virtual labs with subject software for the course (Vlabs), flipped classes (FC), hybrid or blended learning with computer-based communication (CBC), asynchronous learning (AS), and self-work (SW).

- Students can ask questions via email or a Ding Talk group dedicated to the course. Instructors answer questions on weekdays two-three times a week between 5 and 7 p.m. Moscow time.

- Participants are expected to attend all classes. It is required to attend at least 70% of classes. Omissions without a valid reason may affect the final grade.

- Please **inform the student office** if you will be late or unable to come to a class two or more times. If you miss more than 30% of classes without informing the student office, you will not be allowed to the final test.

- **The teacher is not authorized to dismiss students from classes**, give permission for internships, deal with the student's medical documents, etc. All such operations must be carried out by the student through the student office. And then the student office informs the teacher about the passes.

- Participants who successfully complete the attendance and the imperative class work will be allowed to the final test. The success in the final test will determine the result mark for the course.

- Individual DingTalk calls or grouping webinars will be used for debt closing if it is necessary.

Procedure for Counteracting Academic Plagiarism

This section is based on the Regulations on Ongoing Assessment, Interim Certification, and Repeated Interim Certification of ITMO University Students and the ITMO University Student Code. Academic integrity is a fundamental principle of the educational process, and respect for copyright is an element of professional ethics.
Plagiarism is a violation of Clause 3.3.2 of the Student Code and is considered proven if:

- another person’s name, ID, or signature is found in the document;
- the work contains identical structure, wording, or conclusions to other works, and the student is unable to explain or justify the content;

- file metadata indicates borrowing.

Response measures:

- *First case*: warning, notification recorded, student right to explanation; if confirmed — grade 0, work must be redone, grade assigned under standard course rules.
- *Second case*: grade 0, work must be redone, but the maximum grade is limited to the minimum possible for this control point.
- *Third case*: grade 0 without the right to resubmission; the instructor may refuse to check further work on this control point.
  According to Clause 6.59 of the Regulations, if at least one key checkpoint is graded 0, it results in an academic debt (2FX), regardless of the total score.

Procedure for review:
Responsibility lies with the instructor and assistants. Both manual checks and automated systems are used. If plagiarism is confirmed, the instructor must:

- notify the Student and Faculty Office;
- prepare an official report (student's full name, course title, description of violation, supporting evidence).

Course checkpoints

Course has 3 labs (80%) and final test (20%). Lab 1-2 and final test are critical for course (key checkpoints) need to be passed for passing the course. According to Clause 6.59 of the Regulations, if at least one key checkpoint is graded 0, it results in an academic debt (2FX), even if the total score if more or equal 60 points.

Checkpoint	Status	Points
Lab1	Key	20
Lab2	Key	30
Lab3	Optional	30
Final test	Key	20
Summary	100	

Grade System

Below is the breakdown of grades for the course.

The result mark of the course is made up of scores for lab assignments (max 80 points) and the final test (max 20 points).

Participants with 60 points or more **AND** passed three lab assignments and final test for more than 0 scores, will be marked as successfully accomplished the course.

The description of the result mark and grade is presented below:

Result score (%)	Result grade	Result mark	Explanation
60 – 100	P	Pass	Passed
0 – 59	F	Fail	Not passed

Course schedule

№	Topic	Class Type	Hours	Date	Time	Group	Teacher	Comments
1	Introduction and general concepts	Lecture	1	28.09.2026	13:30-15:05	AT4	Alexey Vedyakov	
2	Approximation and identification	Lecture	1					
3	Regression	Lecture	1	29.09.2026	13:30-15:05	AT4	Alexey Vedyakov	
4	Least Squares Estimate (LSE) I	Lecture	1					
5	MATLAB and Identification	Lab	1	09.10.2026	13:30-15:05	AT4	Alexey Vedyakov	
6	Approximation	Lab	1					
7	Least Squares Estimate (LSE) II	Lecture	1	12.10.2026	13:30-15:05	AT4	Alexey Vedyakov	
8	Least Squares Estimate (LSE) II	Lecture	1					
9	Least Squares Estimate (LSE) III	Lecture	1	13.10.2026	13:30-15:05	AT4	Alexey Vedyakov	
10	Least Squares Estimate (LSE) IV	Lecture	1					
11 12	LSE practice	Lab	**3**	14.10.2026	13:30-15:05	AT4	Alexey Vedyakov	
13	Online identification	Lecture	1	20.10.2026	**10:00 – 12:25**	AT4	Alexey Vedyakov	#108
14	Gradient descent method I	Lecture	1					
15	Gradient descent method II	Lecture	1					
16	Parameters estimation	Lab	2	22.10.2026	08:05 – 09:40	AT4	Alexey Vedyakov Alexey Ovcharov	#201 + #203
17	LSE and Optimization							
18	LSE parameter estimates	Lab	**3**	23.10.2026	**10:00 – 12:25**	AT4	Alexey Vedyakov Alexey Ovcharov	#305 + #307
19	LSE parameter estimates							
20	Gradient descent loss function							
21 22 23	Dynamic Regressor Extension and Mixing (DREM)	Lecture	**3**	27.10.2026	**10:00 – 12:25**	AT4	Alexey Vedyakov	#301 + #303
24 25	Implementation of gradient descent method	Lab	2	28.10.2026	10:00 – 11:35	AT4	Alexey Vedyakov Alexey Ovcharov	#305 + #307
	Questions, Consultation		1					
26 27	LSE for online identification	Lecture	2	29.10.2026	08:05 – 09:40	AT4	Alexey Vedyakov	#108
28	Online identification of dynamical models	Lab	**3**	29.10.2026	**10:00 – 12:25**	AT4	Alexey Vedyakov Alexey Ovcharov	#305 + #307
29	Implementation of gradient descent method							
30	Implementation of gradient descent method							
31	Online identification of dynamical models	Lab	1	30.10.2026	08:05 – 09:40	AT4	Alexey Vedyakov	#201 + #203
32	DREM example	Lab	1					
33	Online identification and excitation	Lab	1	06.11.2026	13:30 – 15:05	AT4	Alexey Vedyakov Alexey Ovcharov	
34	Online LSE implementation	Lab	1					
35	Comparison of estimators	Lab	1	09.11.2026	13:30 – 15:05	AT4	Alexey Vedyakov Alexey Ovcharov	
36	MATLAB Identification tools	Lab	1					

№	Topic	Class Type	Hours	Date	Time	Group	Teacher	Comments
37 38	Predictive model	Lab	2	11.11.2026	13:30 – 15:05	AT4	Alexey Vedyakov Alexey Ovcharov	
39 40	Questions, Consultation	Lab	2	16.11.2026	13:30 – 15:05	AT4	Alexey Vedyakov Alexey Ovcharov	
	Final test	Test	2	18.11.2026	13:30 – 15:05	AT4	Alexey Vedyakov Alexey Ovcharov	